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Application of machine learning algorithms for solar power forecasting in Sri Lanka

P. A. G. M. Amarasinghe
Department of Electrical Engineering
University of Moratuwa
Katubedda, Sri Lanka
gihan071@hotmail.com

S. K. Abeygunawardane
Department of Electrical Engineering
University of Moratuwa
Katubedda, Sri Lanka
sarangaa@uom.lk

***Abstract*— Reliability and stability of a power system get decrease with the integration of large proportion of renewable energy. Renewable sources such as solar and wind are highly intermittent, and it is difficult to maintain system stability with intolerable proportion of renewable energy injection. Solar power forecasting can be used to improve system stability by providing approximated future power generation to system control engineers and it will facilitate dispatch of hydro power plants in an optimum way. Machine Learning (ML) algorithms have shown great performance in time series forecasting and hence can be used to forecast power using weather parameters as model inputs. This paper presents the application of several ML algorithms for solar power forecasting in Buruthakanda solar park situated in Hambantota, Sri Lanka. The forecasting performance of implemented ML algorithms is compared with Smart Persistence (SP) method and the research shows that the ML models outperforms SP model.**

Keywords—solar power forecasting, renewable energy, solar power in Sri Lanka, machine learning for forecasting

I. INTRODUCTION

The Buruthakanda solar park is the first commercial-scale solar power station in Sri Lanka which is owned and operated by the Sri Lanka Sustainable Energy Authority. During years 2015, 2016 and 2017, solar power injection to the power grid has been increased exponentially. Sagasolar power station (10MW), Laughs power station (20MW) and Welikanda solar park (10MW) are some commercial scale solar power stations which were built during last three years. Battle for solar energy is a new community-based power generation project which promotes the setting up of small solar parks on rooftops and it is expected to add 200MW of solar electricity to national grid by 2020 [1]. In Sri Lanka, renewable energy technologies will become dominant as it will exceed the major hydro capacity by 2020. Even though the solar has an equal capacity share as the wind, its energy share will take the 3rd place owing to the lower plant factor (~16%) [1].

Unlike other renewable energy sources, solar power is highly intermittent and huge supply power variation will cause power system to be unstable. The intermittency of the solar energy limits the amount of solar power injection to the power grid to maintain the power system stability and reliability. An accurate solar power forecast can be used to efficiently dispatch hydro power stations according to the future solar power generation. Also, the operation of costly diesel generators can be minimized, and pump storage plants will not be necessary since the solar power can be managed effectively knowing the future power generation.

ML uses statistical techniques to give computers the ability to “learn” with data, without being explicitly programmed. Mainly, there are two application categories of ML; regression and classification. The application of solar power forecasting requires regression techniques. Artificial Neural Networks (ANNs), Support Vector Machine Regression (SVMR), and Random Forest (RF) regression are some of the ML regression techniques which can be used for time series forecasting. Electrical power output of a solar Photo Voltaic (PV) panel depends on weather parameters and physical parameters. Solar irradiance, cloud cover, humidity and environment temperature can be considered as main weather parameters that affect the solar power generation. The physical parameters that affect the solar power generation are dust particles on the panel and panel degradation. ML algorithms can be implemented using predicted weather parameters as model inputs and solar power forecast as the model output. The ML algorithm adapts to physical parameters due to its continuous training nature.

Time series forecasting using ML is an active research field. Especially, solar power forecasting is a highly demanded research work due to its importance in generation dispatching. Several SVMR based prediction models are described in [2, 3]. Nageem and Jayabarathi [4] have proposed SVMR model including correction factors for different weather conditions.

In [5], authors have proposed a new forecasting system which is a hybrid of different models, such as electrical and statistical models. Different Numerical Weather Prediction (NWP) sources are used to obtain weather parameters. It results in a lower RMSE compared to the individual models that run separately. Deep learning models for solar power forecasting is proposed in [6]. Various types of deep neural networks such as Deep Belief Network (DBN), Long Short-Term Memory (LSTM) have implemented and results were compared with a physical model. DBN and auto encoder-based LSTM shows good accuracy with lower root mean squared error. Guo et al. [7] proposed an ensemble learning method to forecast solar power output, combining the state-of-art statistical learning methods. Multiple linear regression model (MLR), Neural Network (NN) model and RF regression model together create the ensemble. In [8], authors have presented separate prediction models for different types of days, where these types are determined using clustering of weather patterns. As prediction models, they use ensembles of neural networks to predict the power output for a given day based on the weather. An improved solar power forecasting algorithm based on ANN model with fuzzy logic pre-processing is proposed in [9]. A three-layer feed forward model is proposed using back-propagation as the neural network training algorithm. An error correction factor based on the previous 5 minutes forecasted output is included to the input layer to minimize the forecast error.

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In this paper, several ML techniques i.e. DBN, SVMR and RF are applied for solar power forecasting. The prediction models are developed for first phase of the Buruthakanda solar park which has a capacity of 737kW. SP model is used as the reference model and forecasting performance of machine learning models are compared with the SP model.

This paper is organized as follows. Section II provides a brief introduction to available solar power forecasting methods. In Section III, performance measures used to evaluate the outcomes are described. Section IV explains the implementation process of the selected ML models. In section V, several case studies are presented. In section VI, outcomes of the implemented ML models are discussed. Conclusions are given in section VII.

II. SOLAR POWER FORECASTING METHODS

There are several methods for solar power forecasting. The simplest method is the persistence or smart persistence model that forecasts future power generation for a short term (2~3 hours) using historical power generation data. This method can be used as a reference model to compare other forecasting techniques. Satellite imaging and sky imaging methods can be used to identify the amount of cloud cover and using cloud movement data they can predict future power generation for a short-term period. ML models can be used to map power generation using future weather parameters as inputs. Therefore, to predict future power output, forecasted weather parameters are required. In addition to that, physical models which are unique to each solar park provide future power generation according to the plant parameters and forecasted irradiance.

A forecast is typically performed in a two-stage process. Firstly, a NWP is generated for a certain time period at a certain location. Then, the generated NWP is used to forecast the power generation using forecasting algorithms. This can be a physical model, a statistical method, or a ML technique. The forecasting horizon of the ML or physical models depends on the required forecasting accuracy. The shorter the forecasting horizon, the better the forecasting accuracy. NWP accuracy drastically reduces with long forecasting horizons.

A. Deep belief network

DBN is a class of deep neural network which has multiple hidden layers. DBNs capture higher-level representations of input features by pre-training ANN weights in an unsupervised fashion [10]. DBNs can model the joint distribution between observed vector x and the l hidden layers h^k as shown in (1) [11].

$$P(x, h^1, \dots, h^l) = \left(\prod_{k=0}^{l-2} P(h^k|h^{k+1})\right)P(h^{l-1}, h^l) \qquad (1)$$

Where $x = h^0$, $P(h^{k-1}|h^k)$ is a conditional distribution for the visible units conditioned on the hidden units of the Restricted Boltzmann Machine (RBM) at level k and $P(h^{l-1}, h^l)$ is the visible-hidden joint distribution in the top-level RBM. This is further illustrated in Fig. 1. Training a DBN is typically done in greedy manner, i.e. the DBN is trained to optimize locally at each layer, rather than training to reach the global optimum.

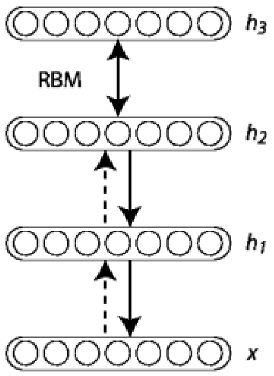

Fig. 1. Stack of Restricted Boltzmann Machines

The first layer is trained as an RBM and it models the input $x = h^0$ as its visible layer. This first layer can be used to obtain inputs to the second layer. Then second layer is trained as RBM taking outputs of the first layer as its inputs (samples or mean activations). Contrastive Divergence (CD) method [12] is used for this initial pre-training process. This process can be continued up to desired number of layers. Final step is fine-tuning each association using gradient descent method.

B. Support vector machine regression

Support Vector Machine (SVM) is a supervised machine learning algorithm which can be used for both classification and regression. The SVMR uses the same principles as the SVM for classification, with only a few differences. The idea of SVMR is based on the computation of a linear regression function in a high dimensional feature space where the input data are mapped via a nonlinear function.

Suppose the given training data for time series forecasting is $\{(x_1, y_1)\dots(x_n, y_n)\} \subset \aleph \times \Re$, where $\aleph$ denotes the space of the input patterns. In ε – support vector regression, the goal is to find a function f(x) that has at most ε deviation from the obtained targets y_i for all the training data and at the same time as flat as possible [13]. The linear function f can be described in the form shown in (2).

$$f(x) = \langle \omega, x \rangle + b \quad \text{with } \omega \in \aleph, b \in \Re \qquad (2)$$

In this, errors are not taken into consideration as long as they are less than ε. However, any deviation larger than this should be minimized. The main idea is to minimize the Euclidian norm as expressed in (3).

$$\text{minimize} \quad \frac{1}{2}\|w\|^2 + c\sum_{i=1}^{l}(\xi_i + \xi_i^*)$$

$$\text{subject to} \begin{cases} y_i - \langle w, x_i \rangle - b \le \varepsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \le \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \ge 0 \end{cases} \qquad (3)$$

Where $\varepsilon_i, \varepsilon_i^*$ are slack variables that allow for some errors. The constant C > 0 determines the trade-off between the flatness of f and the amount up to which deviations larger than ε are tolerated. This optimization is done using Lagrange multipliers. By using the derivatives of standard dualization equation utilizing Lagrange multipliers, with respect to ω, b and ξ, the support vector expansion can be obtained as in (4). Computation of b is done by using Karush-Kuhn-Tucker (KKT) conditions.

$$f(x) = \sum_{i=1}^{l}(\alpha_i - \alpha_i^*)\langle x_i, x \rangle + b \qquad (4)$$

For nonlinear mappings, SVM has a trick called the kernel functions. These are functions which take low dimensional input space and transform it to a higher dimensional space. In other words, it does some extremely complex data transformations, then find out the process to separate the data based on the labels or outputs defined. For nonlinear mappings, ω can be described using (5). Respective support vector expression in (6) can be obtained by minimizing the regularized risk function described in (7) and (8).

$$\omega = \sum_{i=1}^{l}(\alpha_i - \alpha_i^*)\phi(x_i) \tag{5}$$

$$f(x) = \sum_{i=1}^{N}(\alpha_i^* - \alpha_i)K(x_i, x) + b \tag{6}$$

$$\text{where} \quad K(x_i, x)\phi(x_i)\phi(x)$$

$$R_{reg}[f] = \frac{1}{2}\|\omega\|^2 + c\sum_{i=1}^{l} L_\varepsilon(y) \tag{7}$$

$$L_\varepsilon(y) = \begin{cases} 0, & \text{for } |f(x) - y| < \varepsilon \\ |f(x) - y| - \varepsilon, & \text{otherwise} \end{cases} \tag{8}$$

C. *Random forest regression*

Tree based learning algorithms are supervised learning methods which can be used for both classification and regression. These methods can be used to map nonlinear relationships. Methods like decision trees, RF, gradient boosting are being popularly used in all kinds of data science problems.

RF is a tree-based machine learning method capable of performing both regression and classification tasks. It also undertakes dimensional reduction methods, treats missing values, outlier values and other essential steps of data exploration. It is a type of ensemble learning method, where a group of weak models combine to form a powerful model. RFs are trained by the bagging method. Bagging or bootstrap aggregating, consists of randomly sampling subsets of the training data, fitting a model to these smaller data sets, and aggregating the predictions. This method allows several instances to be used repeatedly for the training stage given that the dataset is sampled with a replacement.

D. *Smart persistence model*

The persistence model can be used as the basic reference model for solar power forecasting. It assumes that the current conditions will persist so that,

$$P^*(t + c) = p(t) \tag{11}$$

where $P^*(t)$ represents a solar power prediction from the persistence model, c represents the forecast horizon, and $p(t)$ is the measured solar power generation at time t.

This model assumes that the solar power generation for each time horizon c depends only on the previous measured value $p(t)$, which means that the solar power generation remain invariant between time t and time t + c. In the SP model, the solar power generation for each time horizon c is defined as the mean of the previously measured n solar power generation values in a resolution of c. SP model is defined by (12).

$$P^*(t + c) = \text{mean}\{p(t), p(t - c_1), \ldots, p(t - c_n)\} \tag{12}$$

III. Performance measures

Performance measures are used to evaluate the forecasting performance of different forecasting approaches. The commonly used error metrics for point forecasts are: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), bias and absolute deviation. In this study RMSE, MAE and bias measures are selected for evaluating different forecasting methods.

A. *Root mean square error*

RMSE is the standard deviation of the residuals (prediction errors) as shown in (13). This is a measure of how spread out these residuals are. Since the errors are squared before they are averaged, the RMSE gives a relatively high weight to large errors which means the RMSE is more important when large errors are particularly undesirable.

$$RMSE(x^l, x) = \sqrt{\frac{1}{N} \cdot \sum_{n=1}^{N}(x_n{}^l - x_n)^2} \tag{13}$$

B. *Mean absolute error*

MAE measures the average magnitude of the errors in a set of predictions, without considering their direction. It is the average over the test sample of the absolute differences between prediction and actual observation where all individual differences have equal weights as shown in (14).

$$MAE(x^l, x) = \sqrt{\frac{1}{N} \cdot \sum_{n=1}^{N}|x_n{}^l - x_n|} \tag{14}$$

C. *Bias*

The bias can be used to determine whether power forecast is predicting higher or lower values than the measured power output (15).

$$BIAS(x^l, x) = \frac{1}{N} \cdot \sum_{n=1}^{N}(x_n{}^l - x_n) \tag{15}$$

IV. Implementation

This section explains the implementation procedure of each ML algorithm described in section III-A, B and C. Data preprocessing steps are also explained in more detail.

A. *Implementation tool*

Forecasting model design is done using R [14], a programming language for statistical computing and graphics. R Studio, a free and open-source integrated development environment for R is used as the development environment.

B. *Training dataset*

The data set is obtained from the Buruthakanda solar farm, Hambantota, Sri Lanka with the support of the Sustainable Energy Authority, Sri Lanka. It contains weather and generation data recorded in 2011, 2012 and 2013. Data of each year does not contain all twelve month's weather data. For example, 2013 dataset contains weather data and power generation data from January to August only. The dataset includes 1-minutely data, 10-minutely data, hourly data, daily data, monthly data and yearly data. By using 1-minutely data, a large training set can be obtained, and it supports better visualization of predicted data.

The installed nominal power is 1237 MW. The facility was built in two stages, with installed capacities of 737 kW in the first stage, and of 500 kW in the second stage. The second phase was finished in late 2012. Weather data and power generation data used in this study is obtained from the first stage development of the solar farm. Considering the training time duration of ML models, 80 000 observations in year 2013 are used for training and validation processes.

C. Data normalization

Standardizing the inputs can make training faster and reduce the chances of getting stuck in local optima. The dataset includes diffuse solar irradiance, tilt solar irradiance, global solar irradiance, precipitation, open air temperature and mean wind speed. Since the precipitation values are almost always zero, that weather parameter is not used in this work. These weather parameters are normalized between 0 and 1 using min-max normalization (9).

$$Z_i = \frac{x_i - \mathrm{Min}(x)}{\mathrm{Max}(x) - \mathrm{Min}(x)} \tag{9}$$

Measured power output is normalized using the nominal output capacity of a selected part of the PV facility (10).

$$P_{\mathrm{normalized}} = \frac{P_i}{737} \tag{10}$$

D. DBN design

Neural networks with two hidden layers can represent any function and generally, problems that require two hidden layers are rarely encountered. Number of neurons per hidden layer is determined using trial and error method and thumb rules stated in [15] are used to define boundaries in the trial and error process. Resilient back propagation is used as the fine-tuning algorithm after DBN is trained by CD method. Sigmoid activation function is used in both the initial greedy training process and the fine-tuning process.

E. SVM design

1) Kernel selection

Obviously, the relationship between solar power generation and weather parameters is nonlinear. Thus, suitable kernel should be chosen according to the nonlinear relationship. Kernel transforms a nonlinear space to a linear space. Some of the available kernels are, radial basis kernel, polynomial kernel, linear kernel, hyperbolic tangent kernel, Laplacian kernel, Bessel kernel, etc. Authors have applied all the kernels developed in R and Laplacian kernel gives lowest RMSE for forecasted results. It is a general-purpose kernel and it can be used when there is no prior knowledge about the data.

2) SVM design parameters

As stated in section II-B, epsilon regression is used for the SVMR model. Cost of constraints violation of constraints (this is the 'C'-constant of the regularization term in the Lagrange formulation) is chosen as 10. If the value of C is very small, it will lead to under-fitting and selecting too high values for C will result in SVM to over-fit. Considering these limitations, C was determined using the trial and error method. The epsilon in the insensitive-loss function is also determined in a similar manner ($\varepsilon = 0.1$).

F. RF design

In the RF design, the cross-validation method is used as external resampling method and number of resampling iterations is selected as 10 considering the training time.

G. Night time data filter

The power output of a solar panel mainly depends on the tilt solar irradiance which is the total solar irradiance receives by the tilted solar panel. If the tilt irradiance is zero, obviously there is no power output. Therefore, in the case studies in section VI, power output is taken as zero if the tilt irradiance is zero. This will decrease the RMSE of test data and hence the forecasting accuracy will be increased.

V. CASE STUDIES

Three case studies have performed to analyse forecasting performance of proposed ML models under days with different weather conditions namely, clear (sunny), partly cloudy and overcast (cloudy). 2013 January 30, May 30 and May 31 days have clear, partly cloudy and overcast weather conditions respectively. Corresponding days are filtered out before the training phase of ML models. Figs. 2, 3 and 4 show power vs time graphs for each day. Moreover, Table I shows overall RMSE, MAE and Bias values for the implemented ML models. Table II shows RMSE, MAE and bias values of solar power predictions of each day with different weather conditions.

TABLE I. RMSE, MAE AND BIAS VALUES OF ML MODELS AND SP MODEL PREDICTIONS

Model	RMSE	MAE	Bias
SP	0.13571	0.07175	0.02362
DBN	0.03905	0.01380	0.00024
SVM	0.03434	0.01286	0.00077
RF	0.03327	0.00994	0.00034

TABLE II. RMSE, MAE AND BIAS VALUES OF ML MODELS IMPLEMENTED FOR DIFFERENT WEATHER CONDITIONS

Weather condition	RMSE			MAE			Bias		
	DBN	*SVM*	*RF*	*DBN*	*SVM*	*RF*	*DBN*	*SVM*	*RF*
Clear (Jan 30)	0.01577	0.04412	0.01830	0.00836	0.02094	0.00878	0.00494	-0.00373	0.00111
Partly cloudy (May 30)	0.04622	0.04436	0.04334	0.01786	0.01755	0.01688	-0.00123	-0.00004	-0.00045
Overcast (May 31)	0.02610	0.04010	0.02497	0.00866	0.01629	0.00714	0.00340	-0.00810	0.00283

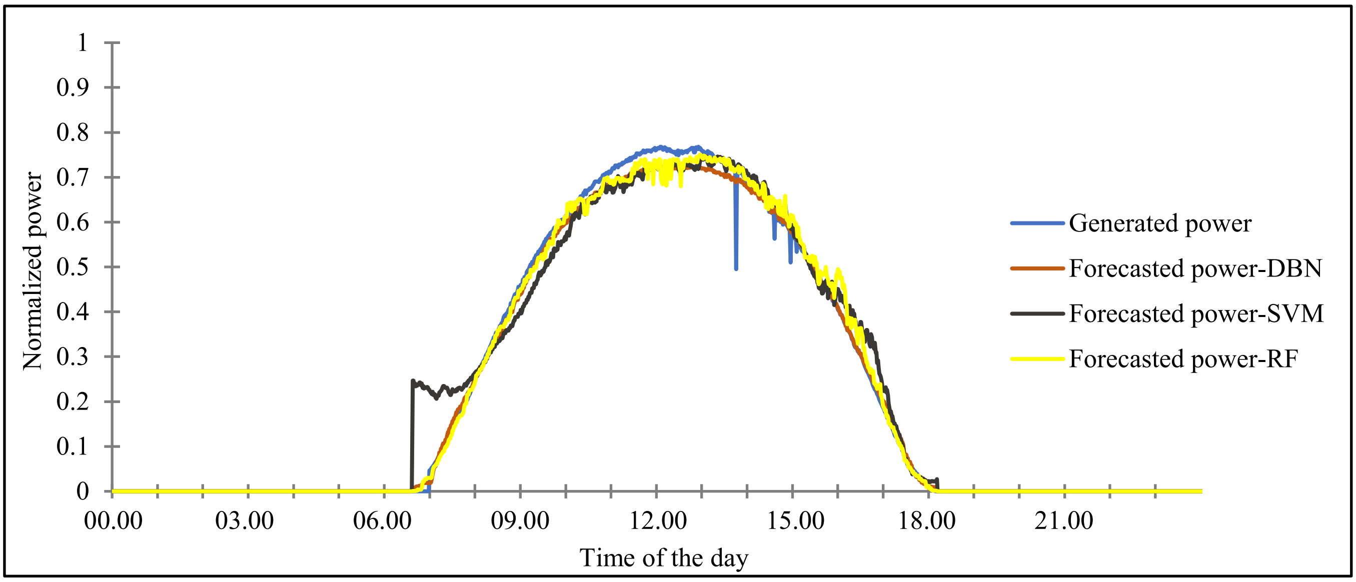

Fig. 2. Actual and forecasted power generation in a clear day. i.e. January 30

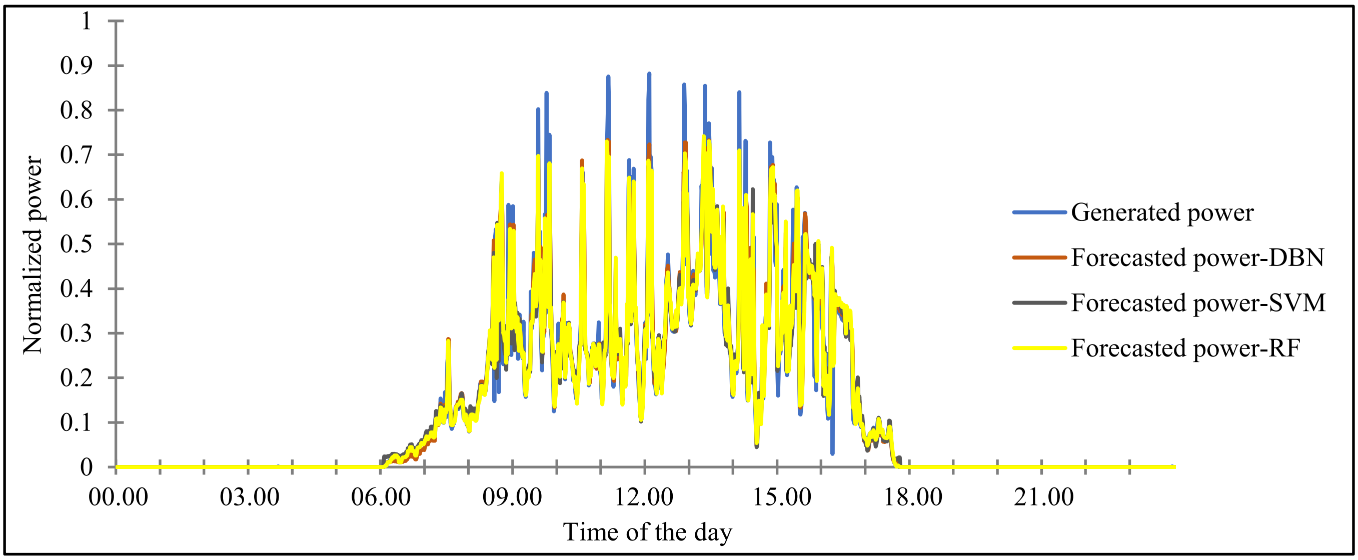

Fig. 3. Actual and forecasted power generation in a partly cloudy day. i.e. May 30

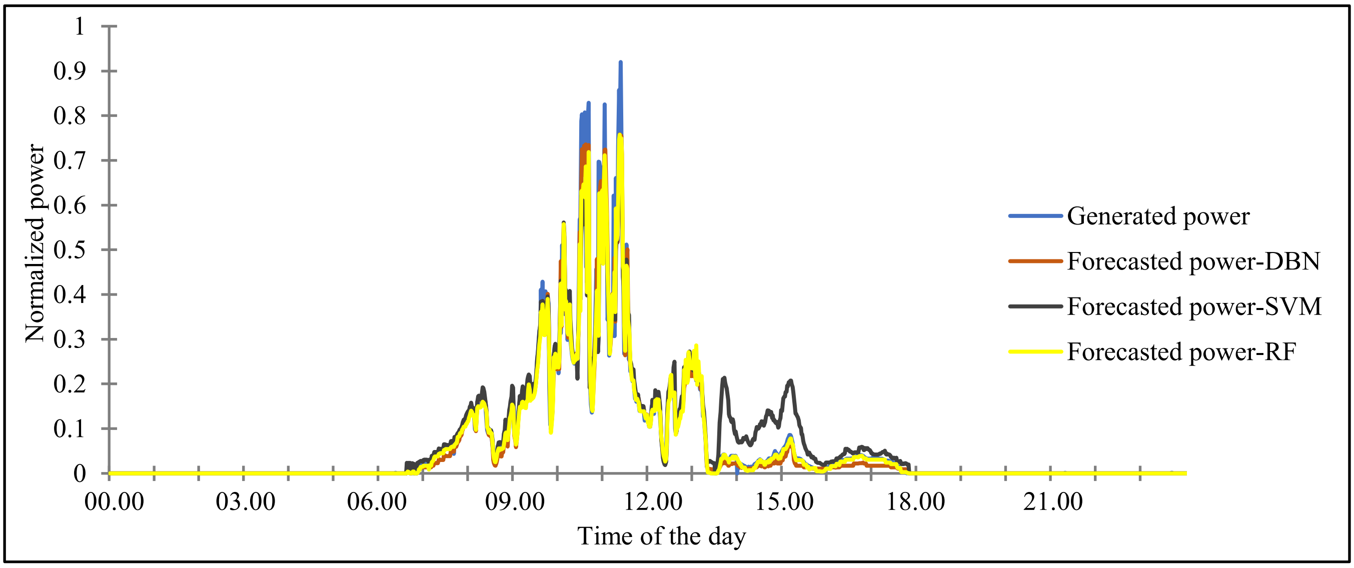

Fig. 4. Actual and forecasted power generation in a overcast day. i.e. May 31

VI. DISCUSSION

As shown in Table I, ML models provide reduced RMSE, MAE and bias values than SP model. RF model provides the lowest RMSE value. RF model is an ensemble model consists of a large number of decision trees and that explains the reason for relatively low RMSE and MAE values. Moreover, bias values of implemented ML models are small and positive. Thus, the forecasted power generation of these models is very close to the actual power generation and the overall power forecast is higher than the actual power forecast by a very slight amount. In addition, low bias values of ML models show that the energy prediction will be very similar to the actual energy generation. The results show that ML models outperform SP model by a huge margin.

According to the results of the case studies shown in Table II, for each ML model, the forecasting performance is higher in the day with clear weather conditions than partly cloudy and overcast weather conditions. The variation of cloud cover is not rapid in clear days and this is the reason for having a less error in power forecasts of clear days. The power forecast of the overcast day has somewhat higher RMSE than that of the clear day. The forecasted power of the partly cloudy day has the largest RMSE and MAE. Forecast error is a function of variability [16]. Partly cloudy conditions have the largest variability and that is the main reason for the above-mentioned differences in forecast errors

The results shown in Table I slightly contradict with the results shown in Table II. As can be seen in Table II, for clear days, DBN has the highest forecasting performance than the other ML models. For partly cloudy and overcast weather conditions RF model gives the best performance. Moreover, DBN and RF models give approximately same forecasting performance for clear, partly cloudy and overcast weather conditions. However, SVM model gives higher RMSE for all weather conditions unlike DBN and RF models. SVM model gives quite larger power output even if the irradiance is very small and that may be the reason for large RMSE values. The variation between the SVM forecast and the actual power generation can be seen in Figs. 2, 3 and 4. It can be seen that the SVM model is not a good forecast model for this dataset.

It may be difficult or impossible to propose a generalized ML model that can be used for any PV plant. The SVM model may give better results for a different dataset even it shows poor performance in this study. The relationship between weather parameters and the generated power may change with the region and the design and construction of the power plant. For example, power forecasting of solar farms with trackers is more difficult than that of conventional solar parks [6]. Sri Lanka is a tropical country and especially, Hambantota district is one of the major dry zones in the country. Therefore, dataset includes more clear data than overcast or partly cloudy data. Due to higher proportion of clear data in the dataset, overall ML model’s performance is higher than the performance of other models implemented for polar regions [6].

VII. CONCLUSION

In this paper authors utilize several ML algorithms for solar power forecasting in Sri Lanka. The forecasting performance of each ML model is compared using several error metrics such as RMSE, MAE and bias. The ML model designs proposed in this paper can better forecast the solar power output than the SP model. It has been shown that ML models such as DBN and RF can be used to forecast solar power with a higher accuracy under clear or overcast conditions. These models can be applied for any forecasting horizon, but the longer the forecasting horizon the greater the NWP error. NWP error is not considered in this study.

This is an ongoing study and this study can be improved further in several ways. The forecasting error can be further reduced by increasing the number of input features: i.e. weather parameters. The model can be improved further by using ensemble of ML models which incorporate the weather type classification. Cloud cover weather parameter is required for this and the RMSE may reduce due to the use of more specific ML model for each weather condition.

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